

Cases of Complexity

The $f(n)$ $\rightarrow$ It gives the running time of the algorithm, depends not only the size n of the input data but also on the particular data.

The complexity function $f(n)$ for certain case x .

- Best Case
- Worst Case
- Average Case

Best Case: The minimum or least possible time required for frequency count of each instruction of an algorithm for implementation.

Worst Case: The maximum time required for frequency count of each instruction of an algorithm for implementation.
* Computational time of an algorithm will never exceed worst case time.

* Computational time for an algorithm will be never be less than best case time.

Average case: It is neither best nor worst. It takes average execution time for frequency count of each instruction of an algorithm for implementation.

Why do we need Asymptotic?

Because

- $\rightarrow$ Different computers have different speeds.
- $\rightarrow$ And also different language take different time -